NCBI Documentation Source Sheet

Internal organisation

NCBI stores genome assemblies in two parallel collections:

- **GenBank**: raw, unfiltered submissions from any group. Accessions start with `GCA_`.
- **RefSeq**: a curated subset of GenBank, re-processed and selected by NCBI staff. Accessions start with `GCF_`.

A given assembly often exists under both forms (a raw GenBank version and its curated RefSeq counterpart), linked to each other.

Data files can be retrieved from https://ftp.ncbi.nlm.nih.gov/genomes/. No login or API key is required to browse or download these files.

Finding assemblies and associated data files

Assemblies are organised by broad taxonomic group, then by species, then by individual assembly.

```
/genomes/refseq/   (or /genomes/genbank/ for the raw collection)
    archaea/
    bacteria/
    fungi/
    invertebrate/
    plant/
    protozoa/
    vertebrate_mammalian/
    vertebrate_other/
    viral/
```

Within each group, one folder per species:

```
vertebrate_mammalian/
    <Genus_species>/
        assembly_summary.txt
        annotation_releases/
        <Accession.version>_<assembly_name>/
```

Example path (*Felis catus*, domestic cat):

```
/genomes/refseq/vertebrate_mammalian/Felis_catus/
    GCF_018350175.1_F.catus_Fca126_mat1.0/
```

The accession is also split into groups of 3 digits to build the full path under `/genomes/all/`, e.g.:

```
https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/018/350/175/
GCF_018350175.1_F.catus_Fca126_mat1.0/
```

This path does not need to be built by hand: it is given directly in the `ftp_path` column of `assembly_summary.txt` (see below).

Inside each assembly folder:

```
<Accession.version>_<assembly_name>_genomic.fna.gz      genome FASTA
<Accession.version>_<assembly_name>_genomic.gff.gz      annotation (GFF3)
<Accession.version>_<assembly_name>_genomic.gtf.gz      annotation (GTF)
<Accession.version>_<assembly_name>_protein.faa.gz      proteins
<Accession.version>_<assembly_name>_rna.fna.gz          RNA / transcripts
<Accession.version>_<assembly_name>_cds_from_genomic.fna.gz   coding sequences
<Accession.version>_<assembly_name>_assembly_report.txt
<Accession.version>_<assembly_name>_assembly_stats.txt
<Accession.version>_<assembly_name>_feature_count.txt.gz
md5checksums.txt
```

Finding statistics files

`assembly_summary.txt` (one per species folder) lists every assembly for that species in a single table, with the main metadata already filled in: accession, taxid, assembly level, submitter, release date, `ftp_path`, genome size, GC%, scaffold/contig count, and gene counts.

It does **not** include N50. N50 (contig-N50, scaffold-N50) is only found inside the per-assembly file `assembly_stats.txt`, together with other details such as sequencing technology and assembly method.

Example, *Felis catus* (GCF_018350175.1):

- Assembly level: Chromosome
- Genome size: 2,425,730,029 bp
- GC content: 42.0%
- Total gene count: 39,307
- Protein-coding gene count: 20,453
- Contig N50: (from `assembly_stats.txt`)
- Sequencing technology: PacBio Sequel 2

Differences between NCBI and GWH

NCBI assembly folders are generally richer than GWH direct submissions: NCBI typically provides GFF and GTF (both formats), an assembly report, a dedicated statistics file with N50, a feature count table and an XML annotation report, in addition to genome/RNA/CDS/protein FASTA files.

GWH direct submissions usually provide a simpler fixed set of files (genome FASTA, GFF, RNA, CDS, proteins, features, MD5) and statistics are shown mainly on the web page rather than in a dedicated stats file.

Internal pipelines

Assembly release (RefSeq)

1. Raw sequencing data and the resulting assembly are submitted to NCBI (GenBank).
2. NCBI evaluates the assembly; a subset meeting NCBI's criteria is selected and curated for RefSeq.

3. A RefSeq accession (`GCF_`) is assigned and linked to the original GenBank accession (`GCA_`).
4. The assembly is annotated (see below) and the files are released under `/genomes/refseq/`.

Annotation pipeline

1. Gene prediction and annotation are run on a given assembly (NCBI's eukaryotic annotation pipeline, or RefSeq prokaryotic pipeline).
2. The result is published as a numbered "Annotation Release" folder.
3. A later annotation release can be produced on the same assembly (new evidence, pipeline update) or on a newer assembly version.

Updates to assemblies and annotations

Assembly versions are indicated by the number after the dot in the accession:

```
accession_core = GCF_018350175
version        = 1
accession      = GCF_018350175.1
```

A new assembly submission for the same effort increases this version number.

Annotation releases are numbered independently of the assembly version, in sequential order, starting from an arbitrary number and increasing by one each time a new release is produced:

```
Felis_catus/annotation_releases/
    104/       (older release)
    105/       (current release)
    current/   (always points to the latest release)
```

The full accession, including the version suffix, should always be kept when referencing an assembly, since different versions of the same accession can correspond to different (though related) assemblies.